

Westly Bouchard

Denver, CO | 720-418-0939

westlybouchard.com | westly.bouchard@gmail.com | linkedin.com/in/wcbouchard

Education

Colorado School of Mines | Golden, CO

Aug 2023 – Expected Dec 2028

- **Master of Science in Robotics** (4+1 Program) | *Expected Completion: Dec 2028*
- **Bachelor of Science in Design Engineering** (Robotics Conc., CS Minor) | *Expected Completion: Dec 2027*
- **GPA:** 3.9 / 4.0 (Dean's List)

Experience

Embedded Software Engineer Intern | Fab2

Sep 2026 – Present

- Developing C firmware, STM32 (CubeHAL), and FreeRTOS packages to support silicon production workflows.
- Designing asynchronous, portable device driver libraries operating over SPI and I2C protocols.
- Building companion desktop debugging apps in Rust using egui to provide rapid prototyping capabilities.

Software Engineer Intern | Freewheel (Beeswax App Team)

Jun 2025 – Aug 2025

- Modernized backend application infrastructure by migrating legacy PHP APIs to Python and Django.
- Developed full-stack features using React.js and TypeScript, integrating data across MySQL, Postgres, and Snowflake.
- Collaborated across agile engineering teams using Git, GitHub, and Scrum to deploy production features.

Computer Science Teaching Assistant | Colorado School of Mines

Jan 2024 – May 2026

- Guided undergraduate engineering students through foundational C++ coursework and object-oriented design.
- Mentored students on low-level memory management, pointers, dynamic allocation, and polymorphism.

Projects

Mossap - Robotics Simulator | C++, WebAssembly, Dear ImGui, VS Code Extension

Jan 2026 – May 2026

- Engineered a lightweight C++ 2D physics engine for differential-drive robots with modular hardware interfaces.
- Compiled simulation runtime to WebAssembly via Emscripten and packaged a zero-config VS Code extension.
- Built interactive telemetry overlays with Dear ImGui to visualize PID trajectory control, occupancy grids, and A* planning.

Swerve Drivetrain | ROS2, C++, Python, Linux, SocketCAN

May 2025 – Dec 2025

- Designed an open-source swerve drive module and built a modular ROS2 control stack to handle kinematics execution.
- Developed a custom C++ `ros2_control` hardware interface to communicate with motor controllers over SocketCAN.
- Implemented non-blocking CAN frame read/write operations utilizing Linux `epoll()` syscalls for deterministic execution.
- Modified open-source motor controller firmware to multiplex dual-encoder signals over CAN bus, reducing wiring complexity.

Skills

- **Languages & Web:** C/C++, Python, TypeScript, Rust, Embedded C, SQL, React.js, WebAssembly, Django, HTML/CSS
- **Systems & Tools:** Linux (SocketCAN, Syscalls), ROS2, Git / GitHub, FreeRTOS, REST APIs, ImGui, HIL Testing
- **Robotics Concepts:** Forward/Inverse Kinematics, PID Control, CAN / SPI Protocols, Microcontrollers, System Architecture